



PROGRAMMING WITH PYTHON

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Python Basics and Standard Library Comprehensions - Set

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PROGRAMMING WITH PYTHON

Set Comprehensions

- Set comprehensions use curly brackets { }.
- { <expr> for <variable> in <iterable> }

```
sentence = "The cat in the hat had two sidekicks, thing one and thing two."  
words = sentence.lower().replace('.', '').replace(',', '').split()  
unique_words = {word for word in words}  
print(unique_words)
```

```
{'in', 'sidekicks', 'thing', 'the', 'hat', 'cat', 'two', 'one', 'and', 'had'}
```



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Set Comprehensions

- Conditional set comprehension
 - `new_set = {expression for variable in iterable if condition == True}`

```
sentence = "The cat in the hat had two sidekicks, thing one and thing two."  
words = sentence.lower().replace('.', '').replace(',', '').split()  
unique_words = {word for word in words if len(word) <= 3}  
print(unique_words)
```

```
{'the', 'cat', 'one', 'hat', 'two', 'and', 'in', 'had'}
```



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Nested Set Comprehensions

- {expression for element in outer_set
- for sub_element in inner_set}

```
sentence = "The cat in the hat had two sidekicks, thing one and thing two."  
words = sentence.lower().replace('.', '').replace(',', '').split()  
vowels = ['a', 'e', 'i', 'o', 'u']  
consonants = {frozenset({letter for letter in word if letter not in vowels}) for word in words}  
print(consonants)
```



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Nested Conditional Set Comprehensions

- {expression for item1 in iterable1
- if condition1
- for item2 in iterable2 if condition2}

```
result = {x * y for x in range(4)
           if x % 2 == 0
           for y in range(5) if y > 2}

result
{0, 8, 6}
```



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Recap

- Set Comprehension
 - Simple
 - Conditional
 - Nested
 - Nested Conditional



THANK YOU

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